Stochastic Processes Semester 1

Extension Questions 5 – Solutions

1. (a) The closed classes are

$$C_1 = \{1, 6\}$$
$$C_2 = \{3, 4\}$$
$$C_3 = \{7\}$$

The remaining states form a single open class.

(b) The limiting distribution starting in state $X_0 = 1$ (and corresponding to closed class C_1) is

$$\pi^{(1)} = (\frac{1}{6},\, 0,\, 0,\, 0,\, 0,\, \frac{5}{6},\, 0)$$

(c) Let $\alpha_2^{(i)}$ be the probability of absorption in class $i = 1, 2, 3$ for a Markov chain that starts from $X_0 = 2$. We see that

$$\alpha_2^{(1)} = \frac{7}{15}$$
$$\alpha_2^{(2)} = \frac{4}{15}$$
$$\alpha_2^{(3)} = \frac{4}{15}$$

(d) The limiting distribution starting from state $X_0 = 2$ is

$$\left(\frac{7}{90},\, 0,\, \frac{16}{135},\, \frac{4}{27},\, 0,\, \frac{7}{18},\, \frac{4}{15}\right)$$

2. (a) This is an irreducible Markov chain which is periodic, with period $d = 4$.

The periodic partition is

$$S_1 = \{1, 6, 7\}$$
$$S_2 = \{5, 8\}$$
$$S_3 = \{2\}$$
$$S_2 = \{3, 4\}$$

(b) i. No
ii. No
iii. No
iv. No
v. Yes
vi. No
vii. Yes

(c) The limiting distribution (when it exists) is

$$\left(\frac{1}{8}, \frac{1}{4}, \frac{1}{12}, \frac{1}{6}, \frac{17}{144}, \frac{1}{24}, \frac{1}{12}, \frac{19}{144}\right)$$

3. (a) The closed classes are

$$C_1 = \{1, 2, 3\}$$
$$C_2 = \{6, 7\}$$

(b) There is no limiting distribution since the closed class C_1 is periodic (and the initial distribution is not spread evenly over the periodic partition).

(c) Let $\alpha_4^{(i)}$ be the probability of absorption in class $i = 1, 2$ for a Markov chain that starts from $X_0 = 4$. We see that

$$\alpha_4^{(1)} = \frac{7}{11}$$
$$\alpha_4^{(2)} = \frac{4}{11}$$

(d) The probability that, starting from state $X_0 = 4$, the chain reaches the set $\{2, 3\}$ before it reaches $\{1\}$, given that it reaches the set $\{1, 2, 3\}$ at all is $\frac{1}{2}$.

(e) The limiting distribution is

$$\left(\frac{7}{22}, \frac{7}{33}, \frac{7}{66}, 0, 0, \frac{4}{33}, \frac{8}{33}\right)$$